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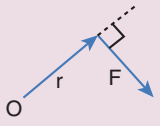
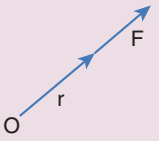
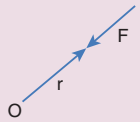
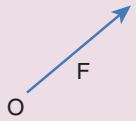
Based on the angle θ between $\vec{r}$ and $\vec{F}$, the torque takes different values.

The torque is maximum when, $\vec{r}$ and $\vec{F}$ are perpendicular to each other. That is when $\theta = 90^{\circ}$ and $\sin 90^{\circ} = 1$, Hence, $\tau_{max} = rF$.

The torque is zero when $\vec{r}$ and $\vec{F}$ are parallel or antiparallel. If parallel, then $\theta = 0^{\circ}$ and $\sin 0^{\circ} = 0$. If antiparallel, then $\theta = 180^{\circ}$ and $\sin 180^{\circ} = 0$. Hence, $\tau = 0$.

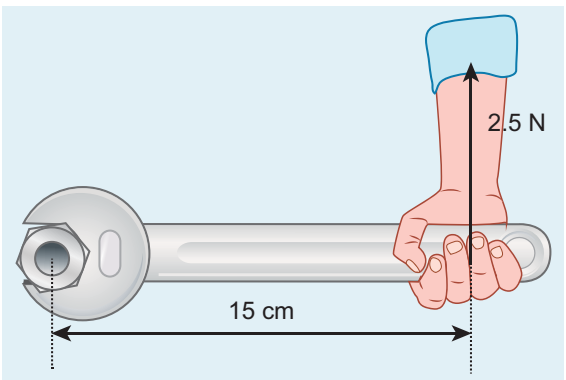
The torque is zero if the force acts at the reference point. i.e. as $\vec{r} = 0$, $\tau = 0$. The different cases discussed are shown in Table 5.1.

Table 5.1 The Value of τ for different cases.

 $\theta = 90^{\circ};\ \tau_{max} = rF$	 $\theta = 0^{\circ}; \tau = 0$
 $\theta = 180^{\circ}; \tau = 0$	 $r = 0; \tau = 0$

EXAMPLE 5.7

If the force applied is perpendicular to the handle of the spanner as shown in the diagram, find the (i) torque exerted by the force about the center of the nut, (ii) direction of torque and (iii) type of rotation caused by the torque about the nut.

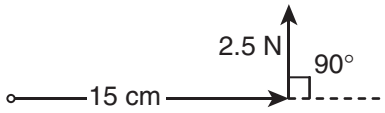


Solution

Arm length of the spanner, r = 15 cm $= 15\times10^{-2}$m

Force, F = 2.5 N

Angle between r and F, $\theta = 90^{\circ}$



(i) Torque, $\tau = rF\ \sin\theta$

$$\tau = 15\times10^{-2}\times2.5\times\sin(90^{\circ})$$

[here, $\sin 90^{\circ} = 1$]

$$\tau = 37.5\times10^{-2}\ \text{N m}$$

(ii) As per the right hand rule, the direction of torque is out of the page.

(iii) The type of rotation caused by the torque is anticlockwise.

EXAMPLE 5.8

A force of $\left(4\hat{i} - 3\hat{j} + 5\hat{k}\right)$ N is applied at a point whose position vector is $\left(7\hat{i} + 4\hat{j} - 2\hat{k}\right)$ m. Find the torque of force about the origin.

Solution

$$\vec{r} = 7\hat{i} + 4\hat{j} - 2\hat{k}$$
$$\vec{F} = 4\hat{i} - 3\hat{j} + 5\hat{k}$$

Torque, $\vec{\tau} = \vec{r} \times \vec{F}$

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & 4 & -2 \\ 4 & -3 & 5 \end{vmatrix}$$

$$\vec{\tau} = \hat{i}(20-6) - \hat{j}(35+8) + \hat{k}(-21-16)$$

$$\vec{\tau} = \left(14\hat{i} - 43\hat{j} - 37\hat{k}\right)\ \text{N m}$$